

The Attralucian Essays:

Exploring the Finite



First Edition

Copyright © 2026 by Kevin R. Haylett. All rights reserved.

This work is shared under the Creative Commons Licence.

Creative Commons CC BY-NC-ND 4.0 License.

<https://creativecommons.org/licenses/by-nc-nd/4.0/>

This work is intended for academic and research use. Any unauthorized distribution, modification, or commercial use beyond the creative use license is strictly prohibited. Typeset in

L^AT_EX

The Attralucian Essays



The World Made Still:
When Verbs Become Nouns in Science

Kevin R. Haylett

The World Made Still:
When Verbs Become Nouns in Science

Overview

Modern science acquired much of its extraordinary power by turning recurrent processes into stable symbolic things. A changing world was divided, named, catalogued, measured and represented mathematically. The noun became one of the principal instruments through which knowledge could travel between observers, institutions and generations. This was not a linguistic mistake. It was a practical achievement of immense consequence. Yet the success of the scientific noun also concealed the compression by which it had been made. At ordinary human scales, the omitted processes often remain slow or stable enough for thing-language to work almost without friction. At the smallest scales, the largest scales and within highly complex systems, however, interaction, transformation, history and registration may become inseparable from what is being described. This essay follows the historical and conceptual trajectory by which processes became scientific things, examines the corresponding role of mathematics and symbolic representation, and proposes that the next foundational task of science is not to abandon nouns, but to recover the verbs from which its nouns were formed.

Entry: A Note Before the Room

Let's start at the beginning. If you are reading this, you are already measuring, and with that measurement you are following a path. My hope is that you will walk with these words and find the path both surprising and rewarding.

Let yourself notice the measurement. You are measuring the light on the page, the shape of the letters, the rhythm of the sentences. And you are measuring against your own history of reading—against books you have loved, the arguments you have trusted, and the words that have carried you through other journeys. You are, perhaps, also measuring whether this text is worthy of your time, and whether it will repay the investment you are making by holding on to the moments of measurement and staying the course.

This is not a metaphor. It is a description of what reading is.

Every act of understanding is an act of comparison. We hold a sequence of symbols in our attention, and we measure them against the sequences we have already encountered. We look for patterns, agreement, and points of connection. We also look for the flaws, the mistakes, the mismatches—what we immediately feel and see as wrong. We build a flowing sense, an internal measurement of what the text is saying. And now you are testing these

words against your own landscape of language—which is also the landscape of your experience.

If the text fits comfortably within your existing framework and locks into place, you will read easily. The words will seem clear, familiar, perhaps even obvious. You may feel that you are being told what you already know. This is how we read.

But if the words are slow, and the text does not fit—if the words are unfamiliar and the assumptions strange—another path begins to open. You begin to feel resistance. Internally, the symbols and words fail to settle into your frames of reference. You may be tempted to stop, to dismiss the text as obscure, pretentious, or simply wrong.

That resistance—this semantic friction—is a signal. It is a sign that you are encountering something that may require you to adjust your measurement. This is the moment when our willingness to take a new path, to allow the reshaping of our internal landscape, becomes decisive. It is the moment we decide if we are open to words we may never have seen before, offering an alternative path of measurement from which we may find new understanding.

The essay presented here is for you to measure. It is about the instruments we use to make measurements—not telescopes or microscopes, but something more fundamental: the nouns by which we divide the world into

stable entities; the 'things' that the nouns point to when we measure.

And so, despite working and knowing that these words, the 'things', and the surrounding text are all text-within-text—a chain of symbols referring to other symbols—the central claim is simple: a noun is not the name of a pre-existing object. It is a halted motion—a dynamical flow that has been compressed into symbolic stability so that it can be carried, shared, and measured.

The word I am about to use—flow, motion, becoming—is itself a noun. I cannot escape that. Nouns are fundamental to our traversal of the landscape of language. But I ask you to hold it lightly, to feel the movement behind the word, and to remember that the naming is not the end of the motion. In measuring nouns, we begin to see that they are not the limit of our understanding.

Yet here is the 'thing': nouns are among our most powerful instruments. They allow us to coordinate action, build institutions, and transmit knowledge across generations. However, when we measure more closely, we find that every instrument has a domain of validity. If we extend noun-language beyond the conditions under which it was formed, we risk mistaking a temporary stabilisation for a permanent feature of the world. Will you notice how the words disappear as you scroll or turn the page? The ink may fade slowly, or the image change in the blink of an eye—but they do change. And we can measure the

change.

If you are prepared to continue, if you are beginning to see value, you will find that what follows traces this trajectory. We will examine how modern science learned to turn dynamical motion into stable things. Then we will look at why that transformation was so successful, and where its limits are now becoming visible. And following the path, we will ask what it means to recover the motion that was compressed into stability—to turn nouns back into verbs when the omitted trajectory, and the dynamics, becomes consequential.

However, before we begin this journey, I ask only this: hold your prior investments lightly. I know they took a lifetime and they make you who you are. That may be a big ask.

Like me, you have spent a lifetime building the frameworks through which you understand the world. Those frameworks are not wrong and have served you well. But they are also measurements—finite, partial, and shaped by the instruments we have learned to accept and use, perhaps without question.

So I present these words as my proxy, for you to measure carefully. They form an invitation to examine one of our most profound instruments: the noun itself. Our goal is not to discard it, but to understand how it works, where it works, and where it may need to be supplemented by

other ways of measuring.

Finally, if you are willing to take that journey, then your room is ready. It is the same room you entered at the beginning—but you may see it differently now.

Let the measurements begin.

A Room Full of Things

Look around the room and name what is there.

There may be a table, a chair, a window, a book, a lamp and perhaps a glass. If another person enters, the two of you can agree upon the inventory with little effort. The table is here. The chair is there. The book lies beside the glass. The language seems so natural that its operation disappears from view. The world appears already divided into things, and the nouns merely arrive to identify what was waiting to be named.

Yet nothing in the room is still in any absolute sense.

The table exchanges heat with the air, bears weight, responds to humidity, accumulates marks and slowly alters chemically. The window transmits some radiation, reflects some, absorbs some and changes dimension with temperature. The lamp is part of an electrical process extending beyond the room. The book is simultaneously paper, ink, manufactured object, commercial object, historical artefact and a symbolic structure awaiting recon-

struction by a reader. The hand that lifts the glass is not a fixed object either. Blood circulates, nerves fire, muscles adjust, cells are repaired and replaced, and the act of grasping changes both the hand and what is grasped.

None of this makes the nouns false. The table is still a table. The glass remains sufficiently glass-like to be lifted. The chair can be sat upon without first giving a complete account of wood, metal, manufacture, mechanical stress and molecular change. The success of the noun rests precisely upon this omission.

A noun retains a distinction while releasing us from the requirement to repeat the complete trajectory of what has been distinguished.

The table can be named, remembered, sold, repaired, inherited, measured and discussed by people who have never occupied the same room. The word travels where the table does not. It carries a compact symbolic stability through time and between observers. The table changes, the observers change and the meaning of the word itself may change, but the noun remains stable enough for the intended work.

This is not a defect added to language after the event. It is one of the primary achievements of language. A world of continuous interaction is made manageable through recurring distinctions. The noun permits us to halt a process symbolically without claiming that the process

has physically stopped.

A noun is a process compressed into symbolic stability.

The formulation is deliberately stronger than the familiar statement that a noun merely names a thing. It asks us to reverse the usual direction of explanation. Rather than assuming that a stable thing exists first and is subsequently given a name, we may ask what processes of distinction, persistence, measurement and convention allow something to be treated as a thing at all.

A room is a useful place to begin because the language of things works exceptionally well there. The objects are neither too small to distinguish nor too large to encounter. Their characteristic rates of change are usually slow compared with the duration of our attention. The room is therefore a domain in which the world and its nouns appear naturally aligned.

The difficulty begins when this local success is treated as a universal grammar of nature.

An Inventory Contains No Motion

Chair. Table. Window. Book. Lamp. Glass.

An inventory contains no motion.

It tells us what has been counted as present, but not how the counted objects arrived, what sustains them,

how they are changing or what relations allowed each to be distinguished as one item rather than another. The inventory is not a complete account of the room. It is a purpose-built compression.

The list removes sequence unless sequence is itself one of the things being listed. It removes most relations unless those relations are explicitly entered. It does not tell us that the book supports the glass, that the lamp illuminates the page, that the open window changes the temperature, or that the observer stands within the same physical process as the objects being observed. The inventory suppresses almost everything in order to retain something useful.

This is why the inventory works.

A complete account of the room would not merely be long. It would be impossible to complete, because the act of describing would itself alter the room and extend the history that required description. Air would move. Light would change. The observer would breathe and shift position. New distinctions would become available while earlier distinctions were being recorded. To communicate anything, the observer must stop.

The stopping is not physical. It is symbolic.

One selects a boundary, an observational window and a degree of permitted variation. Within those conditions, the table is treated as one object, although it contains

many distinguishable parts and participates in many continuing processes. The word *table* is admissible because the omitted changes do not ordinarily prevent the table from performing the role for which it has been named.

The inventory therefore reveals an elementary relation between language and time. What counts as a thing depends partly upon the interval over which it is observed.

Let T_o represent the duration of observation or practical use, and let T_c represent the characteristic duration over which the configuration changes enough to disrupt its classification. Where

$$T_o \ll T_c, \tag{1}$$

noun-like treatment is usually effective. The process changes slowly relative to the observer's window. The omitted dynamics remain present, but they do not materially disturb the intended action.

A chair may be treated as a chair for the duration of sitting upon it. A coastline may be treated as a boundary for a particular map, despite erosion and tidal movement. A species may be treated as a stable category within one investigation, despite evolutionary change and variation between individuals. A word may be treated as carrying a common meaning within one conversation, despite its historical and contextual movement.

A thing, in this practical sense, is not the opposite of a process.

A thing is a process whose transformation is sufficiently slow, recurrent or constrained relative to the observational window.

The distinction between noun and verb is therefore not absolute. It is a distinction of compression, purpose, duration and scale.

The World Made Portable

The ordinary inventory of a room becomes historically consequential when it is expanded into an inventory of nature.

Human societies had named, classified, collected and exchanged things long before the seventeenth century. There was no sudden invention of nouns, specimens or categories. What changed in early modern Europe was the scale and institutional organisation of the work. Observation, collection, measurement, correspondence, demonstration and publication became increasingly coordinated. Distinctions were not merely made; they were prepared for travel.

An observation confined to one encounter could not easily become common scientific knowledge. Another observer needed to recognise what had been seen, or at least recog-

nise the symbolic record of it. The event therefore had to be transformed into a communicable construction: a description, drawing, specimen, measurement, table, diagram or repeatable procedure.

The thing became portable because selected distinctions were detached from the complete trajectory of the encounter.

A plant could be removed from its environment, pressed, labelled and placed within a collection. An animal could be drawn, dissected or preserved. A mineral could be classified by colour, weight, fracture or place of origin. An experimental event could be described as a sequence of operations, observations and results. A phenomenon that occurred once in one room might thereby be re-registered by readers elsewhere.

The world did not literally enter the book. The book carried finite marks from which a reader could reconstruct enough of the original distinctions to participate in the claim.

This is the often-unnoticed achievement behind scientific communication. Knowledge does not travel as the original world. It travels through symbols capable of producing related registrations in another observer.

The seventeenth century enlarged the machinery of this process. Cabinets, catalogues, correspondence networks, experimental reports, illustrated books and learned soci-

eties created a new density of symbolic circulation. The programme associated with Baconian natural history encouraged the accumulation of particulars before the premature construction of universal systems. The Royal Society developed practices of witnessing, reporting, repetition and publication. Hooke's *Micrographia* did not simply show tiny things. It organised instrumentally mediated distinctions into images and names that could be encountered by readers who had never looked through Hooke's instrument. Boyle's experimental histories retained not the entire event but a disciplined symbolic trajectory judged sufficient for another investigator to understand, assess or repeat the work. The *Philosophical Transactions* created a continuing route by which observations could enter institutional memory.

These practices helped make the scientific world cumulative.

The noun was central because the noun could be re-used. A named specimen could recur in different reports. A property could be assigned to an object. A measurement could be attached to a body. A class could contain many instances. A dispute could be organised around whether two observers had encountered the same kind of thing.

The seventeenth-century transformation may therefore be described as a vast programme of symbolic stabilisation. The changing world had to be held still sufficiently for distant observers to share it.

Modern scientific knowledge depended upon making distinctions travel.

The noun became one of the principal vehicles of that travel.

The Scientific Noun

To call the noun a scientific technology is not to suggest that it was consciously designed in the manner of a microscope or pump. Its technological character lies in what it permits.

The noun compresses a potentially unbounded process into a repeatable symbolic unit. It can be entered into an inventory, placed within a hierarchy, joined to a measurement, connected to a property, represented by a variable and carried into calculation. It permits a distinction to persist after the original event has ended.

A living plant becomes a specimen. The specimen becomes a description. The description becomes a class. The class receives a name. The name circulates after the specimen has decayed.

A moving body becomes a trajectory. The trajectory becomes a relation between measured positions and times. The relation becomes a mathematical function. The function can then be analysed in the absence of the body.

A repeated interaction becomes a property. A regularity

becomes a law. A distribution of registrations becomes a particle, state or field. A recurrent form of organisation becomes a system.

At each stage, the language gains stability and loses trajectory.

This loss should not be described too quickly as distortion. Scientific knowledge is possible partly because the whole process is not carried forward. A map omits most of a territory. A specimen omits most of an organism's life. A graph omits most of an experiment. A number omits most of a measurement. A noun omits most of the process it names.

The problem is not that omission occurs. The problem arises when the omitted trajectory becomes necessary to the claim while the compression continues to be treated as complete.

The success of the noun conceals this possibility. Across ordinary scales, the same name repeatedly supports successful action. The table can be used. The species can be identified. The instrument can be built. The prediction can be confirmed within the accepted tolerance. The symbolic construction appears to fit the world, and the route by which it was constructed fades from attention.

The compression worked so well that the existence of the compression could be forgotten.

This is one of the central historical achievements and

risks of modern science. The noun did not merely describe a pre-divided world. It participated in making a common world of stable scientific objects.

Pascal's Unease

Blaise Pascal stands within this transformation rather than outside it. He was a mathematician, experimenter, designer of calculating machinery and participant in seventeenth-century debates concerning pressure, vacuum, proof and method. Yet his writings also preserve a persistent unease about the relation between words, definitions and what they attempt to hold.

The problem appears whenever definition is treated as though it could deliver absolute symbolic security. A word is defined through other words. Those words depend upon prior use, shared recognition or further definition. At some point, language must rely upon terms that are understood without being completely reduced to other terms. The symbolic chain does not terminate by reaching the thing itself. It terminates operationally, within a community capable of continuing the discourse.

This does not make definition useless. Definition constrains use. It narrows ambiguity. It permits a mathematical or scientific argument to proceed. Yet the defined term does not become identical to the process, object or relation to which it is applied.

The word *point* does not lose extension because a mathematical point is defined as having none. The written mark remains extended. The word *vacuum* does not produce an absolute absence. It coordinates a set of experimental and conceptual distinctions. The word *water* does not contain all possible molecular organisations, histories, temperatures, pressures, impurities and uses of water.

Pascal's concern can be read as a warning from within the emerging scientific language: words are necessary instruments, but they cannot become the very things they are used to distinguish.

The warning did not disappear, but it was repeatedly overtaken by success. Scientific terms worked. Definitions enabled calculation. Instruments produced repeatable registrations. Mathematical structures generated predictions. Technologies functioned. Under these conditions, the gap between sign and process could be treated as a secondary philosophical inconvenience.

Yet the gap remained.

A symbol can be reliable without being identical to what it symbolises. A noun can be useful without containing the complete process from which it was formed. A definition can stabilise a term without fixing its meaning for all future contexts.

The scientific revolution did not eliminate this uncer-

tainty. It built a powerful practice capable of operating despite it.

Pascal's Unease

Blaise Pascal stands within this transformation rather than outside it. He was a mathematician, experimenter, designer of calculating machinery and participant in seventeenth-century debates concerning pressure, vacuum, proof and method. Yet his writings also preserve a persistent unease about the relation between words, definitions and what they attempt to hold.

The problem appears whenever definition is treated as though it could deliver absolute symbolic security. A word is defined through other words. Those words depend upon prior use, shared recognition or further definition. At some point, language must rely upon terms that are understood without being completely reduced to other terms. The symbolic chain does not terminate by reaching the thing itself. It terminates operationally, within a community capable of continuing the discourse.

This does not make definition useless. Definition constrains use. It narrows ambiguity. It permits a mathematical or scientific argument to proceed. Yet the defined term does not become identical to the process, object or relation to which it is applied.

The word *point* does not lose extension because a math-

emational point is defined as having none. The written mark remains extended. The word *vacuum* does not produce an absolute absence. It coordinates a set of experimental and conceptual distinctions. The word *water* does not contain all possible molecular organisations, histories, temperatures, pressures, impurities and uses of water.

Pascal's concern can be read as a warning from within the emerging scientific language: words are necessary instruments, but they cannot become the very things they are used to distinguish.

The warning did not disappear, but it was repeatedly overtaken by success. Scientific terms worked. Definitions enabled calculation. Instruments produced repeatable registrations. Mathematical structures generated predictions. Technologies functioned. Under these conditions, the gap between sign and process could be treated as a secondary philosophical inconvenience.

Yet the gap remained.

The Operational Terminus

Pascal's clearest formulation of this gap appears in his unfinished treatise on the geometrical mind. He writes:

french"On ne peut définir toutes les choses,
ni démontrer toutes les propositions... Il faut
donc bien qu'il y ait des mots qu'on entende

sans définition.”

”One cannot define all things, nor demonstrate all propositions... There must therefore be words that are understood without definition.”

The passage is often read as a concession to linguistic imperfection—a recognition that any system of definition must eventually terminate in primitive terms whose meaning is simply *agreed* rather than *derived*. But Pascal’s insight runs deeper than this pragmatic admission.

The words that are ”understood without definition” are not understood because they have somehow reached the thing itself. They are understood because they have reached the *operational terminus* of a particular community’s practice. A child learns what a *point* is not through a formal definition, but through being shown a mark, a tip, or a location. A geometer learns what a *line* is through construction, diagram, and the constraints of demonstration. A natural philosopher learns what *pressure* is through instruments, experiments, and the shared grammar of observation.

Understanding, in this sense, is not the arrival at an unmediated object. It is the capacity to continue the practice within which the word has acquired its force.

The word *point* does not become extensionless because it has been defined as such. The written mark remains

extended. The spoken word occupies time. The concept is carried through a history of use. Yet the community agrees to treat the word as if it could hold that absence. The definition is not a description of a pre-existing ideal object. It is a *symbolic constraint* placed upon future use.

This is what makes Pascal's warning so close to our present argument. He did not conclude that words must be abandoned. He concluded that words must be used with *awareness of their provenance*. The definition is a tool, not a revelation. It coordinates action without containing the complete trajectory from which it was formed.

Pascal's unease therefore anticipates the central distinction of this essay: a symbol can be reliable without being identical to what it symbolises. A noun can be useful without containing the complete process from which it was formed. A definition can stabilise a term without fixing its meaning for all future contexts.

The gap remains. But it is not a failure. It is the *condition of symbolic communication*. The task is not to close the gap, but to remember that it exists—and to return to it whenever the compressed trajectory becomes consequential.

A symbol can be reliable without being identical to what it symbolises.

The scientific revolution did not eliminate this uncer-

tainty. It built a powerful practice capable of operating despite it.

french”On ne peut définir toutes les choses,
ni démontrer toutes les propositions... Il faut
donc bien qu’il y ait des mots qu’on entende
sans définition.”¹

Nouns That Move

It would be a mistake to describe nouns as fixed while verbs alone carry change. A noun is not stable merely because its grammatical form presents something as a thing.

Words move historically.

Their meanings expand, contract and divide. Their permitted uses alter. Different communities may retain the same mark while attaching it to different measurements, instruments and theoretical relations. The noun survives while the scientific object changes beneath it.

The word *atom* offers an obvious example. It carries a historical implication of indivisibility, yet the scientific structures admitted under the word have changed pro-

¹Pascal’s *De l’esprit géométrique* was written around 1655 but published posthumously. It was composed within the same intellectual milieu as the early Royal Society, though Pascal was not a member. His concern with the limits of definition circulated in manuscript form and was known to several figures in the emerging scientific networks of the mid-seventeenth century.

foundly. The same noun has referred across time to substantially different constructions: philosophical minima, chemical units, internally structured systems, probabilistic descriptions and instrumentally reconstructed patterns. The continuity of the word can create an appearance that the same thing has been steadily revealed, when the network of measurements and permissible claims has been repeatedly transformed.

The same applies to *element*, *species*, *force*, *field*, *gene*, *information*, *intelligence*, *memory*, *particle* and *state*. The noun remains recognisable while the relations that give it scientific meaning change.

There are therefore at least two dynamics in every scientific term:

[label=0.]the changing process or set of registrations being described; the changing historical trajectory of the word used to describe it.

These trajectories do not necessarily move together. An inherited noun may persist because institutions, textbooks and mathematical formalisms have stabilised it, even when the underlying measurements have changed. Conversely, the measured process may remain similar while terminology changes because a new discipline, instrument or theoretical system reorganises the symbolic field.

A scientific noun is therefore not a fixed box placed over

a moving world.

A scientific noun is a temporarily stabilised term attached to a temporarily stabilised distinction.

This is close to the historical insight associated with Thomas Kuhn. Scientific change does not always consist of adding new facts to terms whose meanings remain constant. The relations between terms, instruments, exemplars and accepted practices may change together. The same word can occur before and after a major conceptual transition while no longer doing precisely the same work.

The continuity of language can conceal discontinuity in the scientific world.

Our present argument adds a further layer. The scientific world itself is symbolically reconstructed from finite acts of registration. When the network changes, it is not only the meaning of the noun that changes. The conditions under which the distinction is produced may also change. A new instrument may make a new pattern visible. A new computational method may divide a previously continuous distribution into clusters. A new mathematical formalism may permit a different entity to be inferred.

The noun appears to name a discovered thing, but the thing is inseparable from the evolving trajectory by which it became distinguishable.

Why Thing-Language Works

The historical success of noun-language requires explanation. It is not sufficient to point out its limitations. A framework that cannot explain why the existing language works so well has not understood the problem.

Thing-language works because much of human activity occurs within a middle range of scales and durations. Many structures persist long enough, and with sufficiently constrained variation, for their internal dynamics to be ignored during ordinary action.

A stone remains stone-like while it is carried. A house remains house-like while it is occupied. A tree changes continuously, yet its changes are usually slow enough for it to be identified across seasons and years. A human body changes materially, but maintains enough organised continuity to support personal and legal identity. A book persists through many readings. A tool remains usable despite wear.

At this scale, persistence dominates transformation within the observational window.

The observer does not need a complete account of molecular exchange, manufacture, environmental history and future decay in order to use the object. The noun preserves the distinctions relevant to the action and suppresses those that are not.

This suggests a scale-dependent account of scientific language:

Noun-language is most reliable where persistence dominates transformation within the observational window. Process-language becomes increasingly necessary where transformation, relation and registration dominate apparent persistence.

The formulation does not divide reality into true processes and false things. It describes when thing-treatment remains operationally admissible.

A flame is real, but it is not real in the same manner as a stone. A storm is real, but its boundary, material membership and identity depend upon changing organisation. A whirlpool cannot be separated from the moving water that forms it. A melody is not contained in any single note. An organism persists through continuous exchange and replacement. A language exists through repeated acts of use, learning, forgetting and re-registration.

Ordinary grammar already recognises many processes as nouns. We speak of a storm, a flame, a current, a conversation, a life and a history. The grammatical noun does not guarantee material fixity. What matters is the tendency to allow the symbolic stability of the word to produce an expectation of independent and enduring objecthood.

The noun works until the process it suppresses becomes

the subject of the investigation.

Where Things Begin to Move Again

The limits of thing-language become most visible at three boundaries: the very small, the very large and the highly complex.

At very small scales, the scientific entity may be inferred from interactions, traces, deflections, pulses and distributions. The object is not encountered in the same manner as a table. It is reconstructed from registrations produced through instruments and interpreted through mathematics.

At very large scales, the object may exceed any single observational frame. A galaxy, climate system or universe is assembled symbolically from spatially and temporally distributed measurements. The noun promises a unity that no single act of observation supplies.

In highly complex systems, the relevant behaviour may arise through changing relations, feedback, memory and history. An ecosystem is not explained by listing organisms. A language is not a dictionary. A mind is not an inventory of neurons. An economy is not a list of agents. A neural network is not explained by enumerating parameters.

At each boundary, process returns.

This return does not show that the noun was always useless. It shows that the noun had a domain of validity. The trouble begins when a successful middle-scale grammar is extended into domains where the conditions that supported it no longer hold.

The scientific language then performs a familiar move. A recurring pattern is registered. The pattern is given a label. The label becomes a state, property, component or object. The noun then enters explanation as though it had existed independently before the measurements that produced it.

The route can be expressed as follows:

$$\text{interaction} \rightarrow \text{distinction} \rightarrow \text{registration} \rightarrow \text{pattern} \rightarrow \text{classification} \rightarrow$$

(2)

Once the entity has been established, ordinary prose reverses the sequence:

$$\text{entity} \rightarrow \text{property} \rightarrow \text{observed effect.} \quad (3)$$

The entity now appears as the cause of the registrations from which it was inferred. This may be a legitimate explanatory compression, but the circular route must remain recoverable when the distinction is uncertain or scale-dependent.

Registration Before Representation

Before something can be represented, a distinction must be registered.

This elementary sequence is often skipped because scientific language begins with already-formed symbols. The variable is introduced. The object is named. The measurement is entered into a table. The graph is drawn. Yet each of these symbolic constructions depends upon a prior interaction in which some difference became available for retention.

The sequence is:

interaction $\rightarrow$ distinction $\rightarrow$ registration $\rightarrow$ symbolic representation.
(4)

Registration is not passive copying. An instrument, observer or material system responds selectively. Some differences are stabilised; others are lost. A photographic surface, detector, ruler, balance, sensor, nervous system or written language does not retain the complete event. Each has finite resolution, range, geometry, history and cost.

The registered mark is therefore not the event itself. It is a finite consequence of an interaction.

Representation begins when the mark is placed within a

symbolic system. A deflection is assigned a number. A pattern is assigned a name. A measurement is associated with a unit. A cluster is described as a state. A sequence becomes a trajectory. The symbolic construction can then be manipulated, compared and communicated.

When a reader later encounters the symbol, the original event does not occur again. The symbol is re-registered. The reader reconstructs meaning using previous experience, disciplinary training, surrounding language and expectations concerning what may follow.

The complete trajectory is therefore:

$$\text{worldly interaction} \rightarrow \text{registration} \rightarrow \text{symbol} \rightarrow \text{circulation} \rightarrow \text{re-register} \quad (5)$$

The scientific object emerges within this trajectory. It is neither a mere invention nor an unmediated thing delivered intact from the world. It is a stabilised symbolic construction constrained by repeated interactions and shared practices.

The scientific object is what remains after an interaction has been registered, symbolised and stabilised sufficiently to re-enter language.

This formulation restores provenance. It asks not only what the object is called, but how it became available for naming.

The Small Scale: The Thing After the Event

At the smallest scales, the noun often arrives after the event.

An instrument registers a pulse, trace, distribution, collision, deflection or change in state. The registration is not self-interpreting. It is placed within an experimental arrangement, corrected, compared with background, transformed mathematically and assessed against an inherited theoretical structure. Repeated events are gathered into a class, and the class receives a name.

Ordinary language then states that the named entity produced the trace.

This reversal may be scientifically useful. It allows researchers to build models, design instruments and predict new registrations. Yet the language can obscure how dependent the entity is upon the total measurement system.

The question is not whether particles, fields or states are real. Such a question often begins after language has already fixed what kind of answer is permitted. The prior question is more disciplined:

Under what interactions, instruments, symbolic procedures and observational windows does this distinction remain stable enough to be treated as a thing?

This question does not weaken science. It specifies the conditions of the claim.

At small scales, object-language is most secure when different instruments, procedures and representations converge upon a stable distinction. The convergence matters because no single registration carries the complete entity. Repeated and independent trajectories constrain the admissible noun.

Where the distinction changes with instrument, scale, threshold or representation, process-language may be more informative. Instead of asking what the object is in isolation, one asks how the system responds, transforms and becomes distinguishable under specified conditions.

The difference is subtle but foundational. The noun encourages us to imagine a thing bearing properties. The verb encourages us to examine an interaction producing registrations.

The Large Scale: The Object Assembled Across Time

At the largest scales, the difficulty is not that the event is too small to encounter, but that the proposed object is too extended to be observed as one.

A galaxy is reconstructed from radiation arriving at different times, through different instruments and under

changing models of distance, motion and structure. A climate system is assembled from distributed measurements, historical records, proxies, models and classifications. The universe is not observed from outside as a bounded object. Its unity is produced through the coordination of many finite registrations.

The noun compresses this distributed work into a single apparent presence:

2. the galaxy, the climate, the universe.

The unity may be scientifically productive and strongly constrained. It is nevertheless a symbolic achievement rather than a single encounter.

At large scales, histories also exceed observational lifetimes. Processes may unfold over durations for which direct measurement is impossible. The present configuration is used to reconstruct a past trajectory, and the reconstructed trajectory then explains the present configuration. Mathematics and modelling connect registrations separated across time.

Again, the difficulty is not that the reconstruction is illegitimate. The difficulty is that the noun can conceal the distributed provenance through which the object was assembled.

At very large scales, the noun may promise a unity that no single act of observation supplies.

The process must therefore remain visible in the form of provenance, uncertainty, model dependence and admissible alternatives.

Mathematics and the World Made Still

Mathematics is often presented as the language best suited to change. It describes velocity, acceleration, flow, oscillation, growth, decay and transformation. Yet to calculate change, mathematics must first render change into stable symbolic constructions.

A movement becomes a trajectory. A trajectory becomes a function. A changing relation becomes an equation. A local comparison becomes a derivative. A sequence of registrations becomes a data series. A distribution becomes an object that can be operated upon.

Mathematics turns change into something manipulable.

This is indispensable. A calculation cannot retain every physical detail of the process from which its symbols were drawn. Variables must be selected. Units must be fixed. Boundaries must be assumed. Measurements must be entered as finite values. Relations must be expressed in forms that permit repeated symbolic operations.

The danger begins when the mathematical object is returned to the world without restoring what was removed

during its construction.

The curve is spoken of as though it were the trajectory itself. The limit becomes a completed value. The state vector becomes the physical state. The probability distribution becomes a property possessed by an object. The equation appears more exact than the registrations from which it was formed.

Mathematics turns change into an object without necessarily turning the world itself into that object.

The distinction can be clarified through geometry.

A mathematical point has no extent. Every written, plotted or displayed point has extent. The symbol does not possess the defining property of the object to which it refers. It is a finite construction through which the definition can be re-registered.

A Euclidean line has no thickness. Every drawn line has thickness, material, irregularity and boundary. A Euclidean circle is defined through ideal relations. Every measurable circumference must be physically instantiated, with finite resolution and provenance.

The word *circle* can therefore refer to several different constructions:

- a formal definition;
- a drawn mark;

- an instrumentally generated boundary;
- a measured circumference;
- a computational representation;
- an ordinary approximately circular object.

Grammar permits these distinct constructions to travel under one noun. Mathematics permits operations defined for the ideal object to guide work upon finite constructions. This is extraordinarily successful, but it does not eliminate the difference between them.

The measurable circumference is not a defective copy of a completed ideal circumference encountered elsewhere. It is the finite circumference available through a particular registration process. Its uncertainty, material extent and method of construction belong to what has been measured.

Mathematical compression remains admissible while those omitted conditions do not materially alter the claim. When they do, provenance must return.

A compressed symbolic trajectory should not be returned to the world as though nothing had been removed.

When the Inventory Fails

Complex systems expose the limits of explanation by inventory.

An ecosystem is not explained by listing its organisms. A language is not explained by listing its words. A mind is not explained by listing neurons. A society is not explained by listing individuals. A neural network is not explained by listing parameters.

The components matter, but the behaviour under investigation occurs through changing relations. Feedback, sequence, memory, delay, constraint and history are not secondary decorations added to otherwise complete things. They participate in forming the apparent things.

A storm is not a substance placed within the atmosphere. It is an organised atmospheric process. A flame is maintained through continuing interaction. A whirlpool persists because water continues to move through a constrained geometry. An organism maintains identity while matter passes through it. A conversation exists only through a sequence in which each contribution changes the conditions for what may follow.

In such systems, what something is may depend upon what it is doing, what it has done and what it can do next.

Identity becomes historical.

The noun still has a role. We need to say *storm*, *organism*, *language*, *memory* and *model*. Yet these nouns should function as access points to processes rather than substitutes for them.

Memory offers a clear example. Ordinary language treats memory as a stored thing. We have a memory, retrieve a memory, lose a memory or place something into memory. The grammar suggests an object kept in a location.

Remembering, however, is an activity in the present. A past event does not remain unchanged inside the system. Traces, habits, dispositions and relations influence a new reconstruction. Different forms of memory persist at different rates. Immediate context, medium-term adaptation and long-term consolidation need not be the same kind of storage at different sizes. They may be interacting dynamical layers.

The noun *memory* is useful, but it can conceal the process by which prior trajectories alter present possibilities.

The same is true of meaning. A dictionary contains words, but meaning does not reside completely inside isolated entries. A word is understood through previous uses, surrounding terms, expectations, social practices and the trajectory of the current sentence. The reader is not merely retrieving a fixed object. The reader is reconstructing a constrained path.

The failure of the inventory is therefore also a failure of isolated symbol-language. In complex systems, the relation may carry more explanatory force than the listed components.

Meaning Is Not Contained in the Noun

The noun–verb distinction is the visible entrance to a deeper problem. The central opposition is not between two grammatical classes, but between isolated symbolic objects and functional symbolic trajectories.

A noun does not carry one fixed meaning through every context. The word *field* means something different in agriculture, mathematics, physics, computing and ordinary speech. The word *state* may refer to a political body, a condition, a mathematical description or a physical classification. The word *model* may refer to a person, object, equation, machine or conceptual framework.

Meaning is constrained by what precedes the word, what follows it, who uses it, under what circumstances, against which alternatives and with what expected consequences.

A reader re-registers a mark through a history of earlier registrations. Two readers can therefore encounter the same noun and reconstruct different trajectories. Even one reader may understand the same word differently after acquiring new experience.

The word is not an empty shell, but neither is it a complete container.

Meaning is carried through trajectories of relation, ex-

pectation and use.

This is why a noun can appear stable while its meaning moves. The mark remains recognisable. The network around it changes.

Scientific writing often suppresses this movement by treating technical definitions as complete. Definitions are essential, but they do not abolish context. A term may be formally constrained within an equation and still shift when it crosses into ordinary prose, another discipline or public communication.

This is particularly visible when a measured distinction becomes a popular scientific thing. A statistical cluster becomes a type. A state becomes a substance. A functional response becomes an intelligence. A generated sequence becomes an intention. A recurring interaction becomes a self.

The movement is usually gradual. No single sentence seems to perform the complete transformation. Yet the trajectory of language carries the reader from measured dynamics to stable nouns.

The Image Inside the Compression

My route into this problem did not begin with a historical thesis about nouns and verbs. It began through practical experiments concerning images, compression and lan-

guage models.

JPEG compression provides a simple but profound disturbance. An image is divided, transformed and represented through coefficients. The visible image is no longer present in the encoded structure in the form in which an observer ordinarily sees it. Yet enough organised distinction remains for an image to be reconstructed.

Where, then, is the image while it is compressed?

The usual answer is that the information is contained in the representation. This is operationally adequate, but it can encourage another hidden noun. The *image* is imagined as a thing stored inside the coefficients, waiting to be released.

A more precise description is that the encoded structure retains relations from which a sufficiently similar visual registration can be produced under an agreed decoding process. The image is not sitting inside the file as a tiny picture. It is recoverable through a trajectory of transformation.

When this question is carried into language embeddings, the difficulty becomes more apparent. A vector is said to contain meaning. A model stores knowledge. A neuron detects a concept. An embedding represents a word.

These descriptions are useful, but they invite a spatial metaphor in which semantic things are stored inside computational containers.

The operation of an embedding depends upon its relations to other vectors, its position within a sequence, the architecture through which it moves and the transformations applied to it. Meaning may not be located inside an isolated vector any more than an image is visibly located inside one JPEG coefficient.

The noun *meaning* can conceal a process of reconstruction.

This practical problem opened into a wider one. If an image can persist through a transformed relation, and if meaning can be generated through sequences of vector relations, perhaps representation is better understood dynamically. Perhaps what is retained is not a hidden object, but a geometry capable of supporting reconstruction.

Trajectory and Reconstruction

Takens-style phase-space reconstruction offers a useful image for this alternative.

A measured sequence from a dynamical system may permit reconstruction of aspects of the wider system through delayed relations. The significance does not lie in any isolated observation considered alone. It lies in the geometry formed across observations.

The sequence is not merely a row of containers. It is a

trajectory.

This suggests a different approach to language. A token need not contain a stable parcel of meaning. Its significance can arise through its relation to earlier and later tokens, the delays and constraints of the sequence, the history of the conversation and the possible continuations available from the current position.

Meaning is then less like an object retrieved from storage and more like a path reconstructed through a changing symbolic landscape.

The formulation aligns with ordinary reading. A sentence does not deliver all of its meaning word by word in isolated units. Earlier words alter how later words are received. Later words can revise the interpretation of what came before. Context is not a static container surrounding the sentence. It is an evolving trajectory that constrains interpretation.

The reader participates in the reconstruction. The marks on the page are finite. The meanings attributed to them depend upon prior language, memory, expectation and present purpose. Reading is therefore a form of re-registration.

Meaning may reside less in symbolic objects themselves than in the relations through which one registration leads to another.

This is the basis of the Functional Symbolic Trajectory. A symbolic sequence carries representation, constraint,

uncertainty and provenance through time. Compression removes part of that trajectory. The noun is one of the most powerful compressions within it.

Two Liquids in One Glass

We can now return to a contemporary example.

A recent popular account of water describes the liquid in an ordinary glass as two different liquids at once. The underlying scientific work concerns recurrent molecular organisations, differences in density and order, possible liquid–liquid transitions, thermodynamic conditions and computational classifications. The public sentence transforms this dynamic account into two occupants of the glass.

The trajectory of language is revealing:

measured variation $\rightarrow$ configuration $\rightarrow$ cluster $\rightarrow$ structure $\rightarrow$ state $\rightarrow$
(6)

Each step is understandable. Each may be useful within an appropriate technical context. Taken together, however, they change the kind of claim being made.

Molecules continually reorganise. Researchers distinguish recurring patterns within this process. The patterns are classified as high-density and low-density forms. Under

particular conditions, the distinction may support the language of phases or phase transition. Popular prose then announces that the glass contains two liquids.

The scientific distinction may be robust. The grammatical transformation remains consequential.

A molecule does not carry a label identifying it as a member of one permanent class. Its relation to neighbouring molecules changes. A descriptor is calculated from the configuration. A classification procedure assigns the resulting arrangement to a region of a representational space. The named state is the terminal product of a chain of interaction, simulation, measurement and symbolic choice.

The noun is not simply discovered sitting inside the water.

This does not show that the scientific work is mistaken. It shows that the public language has compressed a dynamical distinction into object-like entities.

A more disciplined statement would say that liquid water exhibits recurrent and distinguishable regimes of molecular organisation whose relative prevalence and stability depend upon temperature, pressure and the method of registration.

That sentence is less dramatic because it retains more of the trajectory.

The water example is not exceptional. It reveals a general habit of scientific language. We observe recurrent dynamics, name the recurrent forms and then allow the names to become things.

A Noun as a Halted Trajectory

We can now state the central formulation more fully.

A noun is a halted trajectory.

The process has not physically stopped. Language has selected a distinction and treated it as sufficiently stable for a purpose.

The formulation can be expanded:

A noun is a process compressed into symbolic stability.

Each term matters.

Process indicates interaction, transformation and history.

Compression indicates selective omission.

Symbolic indicates that the retained distinction has entered a system of finite marks and relations.

Stability indicates that the construction can be re-registered with enough consistency to support communication, calculation or action.

The noun is therefore an operational achievement rather than a metaphysical guarantee.

Its stability may arise from several sources. The measured process may genuinely persist. The observational window may be short. The instrument may register only selected differences. The community may enforce a conventional classification. The mathematical representation may constrain interpretation. Usually several of these operate together.

The question is not whether the noun is simply real or unreal. The question is whether its use is admissible for the claim being made.

The Admissible Noun

An admissible noun preserves enough of its originating trajectory for the intended scientific purpose.

This requires a change in questioning. Instead of asking only what the thing is, we ask:

- What distinctions produced the noun?
- How were those distinctions registered?
- Which instruments and symbolic procedures were involved?
- Over what duration does the classification remain stable?
- At what scale does it apply?

- Which transformations cause the classification to fail?
- What uncertainty accompanies it?
- Which parts of the originating process were omitted?

The same noun may be admissible in one context and misleading in another.

The noun *water* is highly admissible for drinking, transport, plumbing and much of chemistry. The phrase *two liquids* may be admissible within a carefully specified thermodynamic discussion, yet misleading when it suggests two enduring substances occupying an ordinary glass.

The noun *point* is admissible within a formal geometry, but not as a direct description of an ink mark whose extent is being measured. The noun *memory* is admissible as a broad functional term, but misleading when it encourages the belief that an unchanged object is retrieved from storage. The noun *intelligence* may coordinate useful behavioural distinctions, but can become misleading when it is treated as an independently stored substance inside a person or machine.

Admissibility is therefore contextual, historical and scale-dependent.

This does not weaken scientific claims. It strengthens them by restoring the conditions under which they hold.

The scientific noun becomes more trustworthy when its provenance can be recovered.

Recovering the Verb

Science need not abandon its nouns. A language that attempted to carry the complete process behind every term would become unusable. No reader could reconstruct the full physical, historical and symbolic trajectory of every word in every sentence. Compression is necessary.

The task is not to eliminate compression, but to make it reversible when the omitted trajectory becomes consequential.

Scientific language should be able to move in both directions:

$$\text{process} \rightarrow \text{noun} \tag{7}$$

for communication, classification and calculation, and

$$\text{noun} \rightarrow \text{reconstructed process} \tag{8}$$

for explanation, criticism and boundary testing.

The problem is not nounification alone. It is irreversible nounification.

A mature scientific language would permit its stable things

to be unfolded back into the registrations, relations, transformations and uncertainties from which they were formed. Provenance would not be an optional historical note. It would be part of the meaning of the scientific object.

Mathematics would remain indispensable, but mathematical results would return to measurement with their construction conditions visible. Classification would remain indispensable, but categories would retain the trajectories by which their boundaries were established. Instruments would remain extensions of observation, but their selective registrations would not be mistaken for unmediated access to the world.

Process-language would not replace thing-language everywhere. It would become increasingly important where transformation, relation and registration dominate persistence.

The seventeenth-century project made the world portable by stabilising processes into things. The contemporary task is to preserve that achievement while recovering the motion hidden by its success.

Returning to the Room

Look around the room once more.

The table remains a table. The chair remains a chair. The book remains a book. Nothing in the argument re-

quires these distinctions to be abandoned.

What has changed is the reader's awareness of what the nouns accomplish.

The room does not arrive as a completed inventory. Distinctions are made within an ongoing interaction. Boundaries are accepted. Variations are ignored. Histories are compressed. Words are applied. The words can then be carried into memory, conversation, measurement and calculation.

The noun is not merely a label attached to a finished thing. It is a temporary stabilisation within a continuing trajectory.

The table can be treated as a thing because its relevant organisation persists within the observational window. At another scale or under another question, it becomes wood fibres, molecular exchanges, stress distributions, industrial history, economic relation or ecological consequence. None of these descriptions alone exhausts it. Each stabilises a different set of distinctions for a different purpose.

The room is therefore full of both things and processes, not because reality has been divided into two kinds of being, but because language can halt change at different levels of resolution.

Modern science acquired extraordinary power by learning to perform this halt systematically. It named the re-

current forms, entered them into catalogues, represented them mathematically and built institutions through which the symbols could be repeatedly re-registered. The scientific noun made a common world possible.

But the success of the noun also encouraged us to forget the verb.

At ordinary scales, the loss often remains harmless. At small scales, large scales and within complex systems, the missing trajectory begins to matter. The object dissolves into interaction, exceeds the observational frame or depends upon relations unfolding through time.

The next task is not to reject the world of things. It is to understand the conditions under which things can be made from processes without the process being lost.

Science learned to make the changing world intelligible by holding its processes still. Its next foundational task is to remember the motion from which its stable objects were formed.

Or, in its shortest form:

Science progressed by learning to turn verbs into nouns. Its next foundational task may be learning when to turn the nouns back into verbs.

Preliminary Notes for Revision

This first draft is written as a continuous philosophical essay rather than as a fully documented historical monograph. A later revision should deepen and verify the historical sections through direct engagement with primary sources and specialised scholarship. The following trajectories are especially important:

- Baconian natural history and the accumulation of particulars;
- the Royal Society's practices of witnessing, reporting, cataloguing and replication;
- Hooke's use of microscopy, drawing and naming in *Micrographia*;
- Boyle's experimental histories and the symbolic reconstruction of events;
- Pascal on definition, geometrical method, experiment and the limits of language;
- the historical movement of terms such as atom, species, force, field, state and information;
- Kuhn's account of conceptual and linguistic change across scientific transitions;
- the relation between process philosophy and the distinct registration-based argument developed here;

- the mathematical distinction between ideal objects and measurable constructions;
- contemporary case studies in water, particle physics, complex systems and machine learning.

The central claim should remain historically generous. The scientific noun is not presented as a mistake to be corrected, but as a powerful compression whose domain of admissibility must be understood.

Selected Works for Historical Orientation

- Bacon, Francis. *Novum Organum* and works on natural history.
- Boyle, Robert. Experimental histories and writings on the spring of the air.
- Hooke, Robert. *Micrographia*.
- Kuhn, Thomas S. *The Structure of Scientific Revolutions*.
- Pascal, Blaise. *De l'esprit géométrique*, writings on the vacuum, and the *Pensées*.
- Sprat, Thomas. *The History of the Royal Society of London*.
- Royal Society. Early volumes of the *Philosophical*

Foundations

Transactions.